

a subcritical thorium blanket, independently of the FBR fleet, potentially compressing the wait by 20–30 years,” says Ranjan, the co-founder of ASPL Fusion. Gandhinagar-based ASPL is one of the youngest start-ups working on this technology.

India is already contributing ₹745 crore in 2026–27 for a global fusion project—the International Thermonuclear Experimental Reactor—a multi-country endeavour located in France. Prime Minister Narendra Modi had visited the site in Marseille, France, in February 2025.

ROLE OF PRIVATE CAPITAL

Achieving the 100 GW target is expected to require ₹19,000 lakh crore investment, creating significant opportunities across the nuclear value chain, including EPC contractors, heavy engineering companies, component manufacturers, fuel cycle services, and plant O&M providers, according to research by Prabhudas Lilladher.

Alluding to the SHANTI Act, Ranjan says the entry of private capital does not compromise sovereignty. “It complements gov-



“Fusion-fission hybrid technology offers a parallel route... potentially compressing the wait by 20–30 years”

PRABHAT RANJAN
CO-FOUNDER, ASPL FUSION



ernment investment by funding technology demonstration stages that would otherwise compete for space in an already stretched public R&D budget,” says Ranjan.

India’s nuclear value chain spans multiple segments, with power developers like NTPC, Adani, Tata Power, and JSW Energy participating at the ownership level. The equipment and component ecosystem is supported by players such as L&T, BHEL, MTAR, Walchandnagar, and ISGEC. In EPC and engineering, companies like Power Mech, HCC, and Engineers

India play a key role.

The Union Budget 2025–26 allocated ₹20,000 crore for the Nuclear Energy Mission. SMRs are envisioned for diverse uses, including replacing aging coal plants, supplying industrial captive power, supporting off-grid needs, and enabling hydrogen production.

The interest of India Inc is also evident. JSW Energy has already announced construction of a nuclear power plant in the next three-four years. Land acquisition at multiple locations is already being done for a 700 MW reactor. The company is looking at a mix of reactors to decarbonise its operations.

The Adani Group has entered the space through Adani Atomic Energy Ltd. Tata Power has begun site identification for 20–50 MW SMRs and formed a dedicated internal team to track regulatory developments and evaluate project opportunities. Reliance Industries is also exploring sites for a nuclear project.

The civil nuclear race is on among private players to commercialise the technology for power generation. But for now, the journey remains long and arduous. **BT**

| 79

GOING GREEN

CO₂ avoided per million tonnes to produce 1 GW of electricity

RENEWABLE ENERGY SOURCE



SOLAR
1.1



BIO
1.1



WIND
1.4



HYDRO
2.6



NUCLEAR
1.1

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